

UCT642: ENTERPRISE SYSTEMS

L	T	P	Cr
2	0	2	3.0

Course Objectives: To provide an understanding of skills in understanding architecture and non-functional requirements in developing Enterprise system development and their deployment.

Introduction to Modern Enterprise Systems: Introduction to enterprise systems. Elements of enterprise systems – Business Information system, Decision support systems, Knowledge management systems, Financial and human resource systems. Kinds of Enterprise systems- B2C and B2B models.

Components of Enterprise systems: Channels (Mobile, web, desktop, partner integration), Data management, workflow, Controlling and Auditing, Accounting etc.

Sample Enterprise systems: ERP, SCM, CRM, Product Life cycle management (PLM), HR Systems (HRM), GL systems.

Key characteristics Enterprise systems: Distributivity, Managed redundancy, Exception processing, Collaboration, Data transformation.

Enterprise System architectures: Batch processing, Monolithic, client server, ecommerce, service oriented, microservice, and cloud architectures.

Introduction to Enterprise Application architectures: Layer Architecture, Event driven Architecture, Service oriented Architecture, Microservice architecture, Plug-in architecture

Application architecture Patterns: Layering, Organizing domain logic, Mapping to database, Web Presentation, Concurrency.

Enterprise Application Integration: Introduction to Enterprise Integration, different integration styles. Elements of messaging-based Integration.

Enterprise Integration patterns: Modern service integration techniques. Introduction to WSDL, SOAP. Introduction Restful webservices integration. Differences between SOAP and REST.

Deployment of Enterprise applications: Key requirements in deployment - Stability, capacity, Security, availability, Network, Availability, and Transparency

Concepts of Cloud computing, cloud platforms and their role in Enterprise systems: Core Concepts – Types of Cloud: Private, public, and Hybrid clouds. Advantage of cloud computing – Scaling, Availability, and cost. Disadvantages – Technology overload, Security, Monitoring and troubleshooting, Testing, Latency etc. Cloud service models: - Infrastructure, platform, Software as a Service in Cloud Computing. Major public clouds: Google cloud, AWS, Azure.

Application development and deployment in cloud – Dockers, micro services, Kubernetes, Serverless. Continuous Integration/Continuous Delivery

Introduction to Enterprise Architecture: Importance of Enterprise Architecture. Enterprise architecture models. Zachman Framework, TOGAF Framework

Laboratory Work

Identify an open-source application and ask students to study/develop/modify various elements of the application from enterprise systems architecture, development and deployment perspective. Enterprise Architecture Case studies.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

Upon completion of this course, students will be able to:

1. Understand basic elements of Enterprise systems
2. Develop skills in understanding architecture and non-functional requirements in developing Enterprise system development and their deployment
3. Understand the modern Enterprise Systems.
4. Comprehend the concepts of cloud computing in Enterprise Systems
5. Illustration of requirements to design an Enterprise Application using Open-source tools.
6. To develop critical thinking for the understanding of success and failures of Enterprise adoption by various organizations
7. Understand future trends in Enterprise architectures.

Text Books:

1. Ralph Stair, George Reynold, "Principle of Information Systems", 10th edition.
2. Martin Fowler et al, "Pattern of Enterprise Application Architecture", Addison-Wesley, 2012
3. Gregor Hohpe, Bobby Woolf, Enterprise Integration Patterns: Designing, Building, and Deploying Messaging Solutions,
4. Mark Richards, Software Architecture patterns, 2015, O'Reilly.
5. Sam Newman, "Building Microservices", 2015, O'Reilly.
6. David Farley, Jez Humble, "Continuous Delivery: Reliable Software Releases through Build, Test, and Deployment Automation", Jan 2016

Reference Books:

1. Brendan Burns, Designing Distributed Systems, O'Reilly, 2016
2. Enterprise Integration Patterns - Messaging Patterns Overview
3. Software architecture in Practice 3rd Edition- 2014